

KEMEROVO, Russian Federation

WMO: 296420

Lat: **55.270N**Lon: **86.107E**Elev: **263**StdP: **98.21**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-31.0	-36.8	0.1	-34.0	-33.8	0.2	-30.9	12.0	-9.4	10.5	-9.9	2.7	160	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.0	19.1	27.1	18.3	25.7	17.6	20.7	26.1	19.7	24.8	18.7	23.8	3.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.2	23.1	17.9	13.3	22.2	16.9	12.4	21.3	60.9	26.1	57.2	24.9	54.3	23.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.3	-36.1	31.9	4.4	1.4	-39.2	32.9	-41.8	33.7	-44.3	34.5	-47.4	35.5		
(5)			-35.9	22.7	4.4	1.6	-39.1	23.9	-41.7	24.8	-44.2	25.7	-47.4	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-17.0	-13.9	-5.8	2.8	10.9	17.0	19.1	16.3	9.8	2.4	-7.6	-14.0
	DBStd	13.86	8.01	7.05	6.03	5.51	5.63	4.44	3.29	3.72	4.53	5.33	7.69	7.76
	HDD10.0	3854	836	669	491	224	59	4	0	2	59	239	527	744
	HDD18.3	6168	1094	902	749	467	236	76	32	83	257	493	777	1002
	CDD10.0	847	0	0	0	7	88	214	281	198	54	5	0	0
	CDD18.3	119	0	0	0	0	7	35	54	22	1	0	0	0
	CDH23.3	944	0	0	0	1	94	295	373	163	18	0	0	0
	CDH26.7	202	0	0	0	0	21	64	80	34	3	0	0	0
Wind	WSAvg	4.1	4.1	4.1	4.5	4.5	4.6	3.6	3.2	3.4	3.8	4.3	4.5	4.5
Precipitation	PrecAvg	557	30	25	26	28	43	68	76	68	49	49	49	46
	PrecMax	699	57	57	65	57	92	130	124	146	93	85	74	78
	PrecMin	401	3	1	5	3	9	26	9	29	11	27	20	9
	PrecStd	73	14	12	15	13	20	28	29	26	22	14	14	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	1.2	8.0	20.9	29.2	30.9	31.1	30.1	26.1	18.8	8.2	1.9
		MCWB	-0.1	-0.3	3.8	11.1	16.3	18.5	20.4	19.9	16.7	10.8	5.6	0.5
	2%	DB	-1.8	-0.9	5.2	17.1	25.9	28.5	29.0	27.1	22.0	15.1	5.9	-0.1
		MCWB	-2.6	-1.9	1.9	9.4	14.3	18.2	19.9	19.1	14.3	9.3	4.0	-0.9
	5%	DB	-3.8	-2.1	3.3	14.1	23.2	26.8	27.1	25.0	19.6	12.4	3.2	-2.0
		MCWB	-4.4	-3.0	1.0	8.1	13.2	17.6	19.2	17.7	13.4	7.9	1.8	-2.7
	10%	DB	-5.9	-4.1	2.0	11.1	20.9	24.9	25.2	23.0	17.0	9.9	1.3	-4.0
		MCWB	-6.7	-4.9	0.1	6.2	12.2	17.0	18.4	16.9	12.1	6.8	0.2	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.3	4.3	12.2	17.3	21.0	22.3	21.1	17.5	11.9	6.6	0.5
		MCDB	0.9	0.9	7.4	18.5	26.6	27.1	28.0	27.4	23.9	16.3	7.6	1.1
	2%	WB	-2.5	-1.8	2.5	10.2	15.7	19.6	21.2	19.7	15.3	10.0	3.9	-1.0
		MCDB	-1.8	-1.1	4.4	16.3	23.3	25.8	26.4	25.1	20.6	14.3	5.2	-0.5
	5%	WB	-4.3	-3.2	1.3	8.5	14.2	18.6	20.2	18.7	13.9	8.5	1.9	-2.9
		MCDB	-3.7	-2.4	3.0	13.4	21.2	24.7	25.0	23.8	18.6	12.0	3.2	-2.2
	10%	WB	-6.6	-5.0	0.1	6.5	12.8	17.6	19.4	17.6	12.6	6.9	0.3	-4.6
		MCDB	-5.9	-4.3	1.8	10.5	19.2	23.4	24.0	22.2	16.3	9.4	1.2	-4.0
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.3	7.9	8.7	10.8	9.9	9.4	9.4	9.0	7.2	6.0	6.7
		MCDBR	7.2	7.8	8.3	13.1	14.4	12.7	11.6	12.2	13.1	11.6	6.6	7.5
		MCWBR	7.1	7.4	6.2	7.9	7.0	5.7	5.3	5.7	6.6	7.0	5.4	7.3
	5% WB	MCDBR	7.1	7.8	7.4	12.2	12.9	10.9	10.2	10.9	11.6	10.8	6.4	7.5
		MCWBR	7.1	7.5	6.1	7.9	6.9	5.4	5.3	5.6	6.4	6.8	5.4	7.4
Clear-Sky Solar Irradiance	taub	0.292	0.326	0.375	0.402	0.406	0.403	0.429	0.406	0.371	0.355	0.326	0.296	
	taud	2.008	1.962	1.904	2.042	2.207	2.301	2.229	2.275	2.319	2.222	2.025	1.952	
	Ebn at Noon	585	705	761	810	836	842	809	804	782	683	527	465	
	Edh at Noon	70	107	145	146	131	121	128	115	96	83	69	59	
All-Sky Solar Radiation	RadAvg	0.88	1.89	3.24	4.41	5.05	5.89	5.33	4.37	2.90	1.47	0.73	0.55	
	RadStd	0.10	0.14	0.30	0.28	0.58	0.51	0.38	0.34	0.36	0.17	0.13	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page